The Bootstrap

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This page reviews the bootstrap, a resampling method for standard errors and confidence intervals that does not need a formula for the sampling distribution of a statistic. Its HERS example bootstraps the slope of a simple linear regression. This page is adapted from Vittinghoff et al. [1], Section 3.6.

The HERS data

The examples on this page use the HERS data, which the Comparing Means page describes. The `rmb` R package includes the dataset; `haven::as_factor()` converts its Stata value labels to factors:

```
hers <- rmb::hers |> haven::as_factor()
hers |> head()
#> # A tibble: 6 x 37
#>   HT      age raceth nonwhite smoking drinkany exercise physact globrat poorfair
#>   <fct> <dbl> <fct>  <fct>    <fct>   <fct>    <fct>    <fct>   <fct>   <fct>
#> 1 plac~    70 Afric~ yes      no      no       no       much m~ good    no
#> 2 plac~    62 Afric~ yes      no      no       no       much l~ good    no
#> 3 horm~    69 White  no       no      no       no       about ~ good    no
#> 4 plac~    64 White  no       yes     yes      no       much l~ good    no
#> 5 plac~    65 White  no       no      no       no       somewh~ good    no
#> 6 horm~    68 Afric~ yes      no      yes      no       about ~ good    no
#> # i 27 more variables: medcond <dbl>, htnmeds <fct>, statins <fct>,
#> #   diabetes <fct>, dmpills <fct>, insulin <fct>, weight <dbl>, BMI <dbl>,
#> #   waist <dbl>, WHR <dbl>, glucose <dbl>, weight1 <dbl>, BMI1 <dbl>,
#> #   waist1 <dbl>, WHR1 <dbl>, glucose1 <dbl>, tchol <dbl>, LDL <dbl>,
#> #   HDL <dbl>, TG <dbl>, tchol1 <dbl>, LDL1 <dbl>, HDL1 <dbl>, TG1 <dbl>,
#> #   SBP <dbl>, DBP <dbl>, age10 <dbl>
```

Bootstrap confidence intervals

When to use the bootstrap

The bootstrap [2], [3] estimates the sampling distribution of a statistic by resampling the observed data. It gives standard errors and confidence intervals in three situations where the usual formulas fall short [1, Ch. 3]:

- approximate methods for valid confidence intervals exist, but standard software does not implement them conveniently;
- no closed-form approximate method has been found;
- the data violate the assumptions of the established methods badly enough that their confidence intervals would be unreliable.

The bootstrap procedure

Definition 0.1 (Bootstrap sample)**.** A **bootstrap sample** from observed data $x_1, \dots, x_n$ is a sample of size n drawn with replacement from $\{x_1, \dots, x_n\}$, each draw choosing each of the n observations with probability $1/n$.
Source: Wikipedia contributors [4].

Remark (Repeated observations in a bootstrap sample)

Remark 0.1 (Repeated observations in a bootstrap sample). Because the draws are with replacement, a bootstrap sample usually contains some observations more than once and omits others. For example, a bootstrap sample from the data $1, 2, 3$ might be $2, 2, 3$.

Example 0.1 (Bootstrap samples of five glucose values)**.** Take the baseline fasting glucose values of the first five HERS participants, and draw two bootstrap samples from them:

```r
glucose5 <- hers$glucose[1:5]
glucose5
#> [1]  84 111 114  94 101
set.seed(1)
list(
  sample_1 = sample(glucose5, replace = TRUE),
  sample_2 = sample(glucose5, replace = TRUE)
)
#> $sample_1
#> [1]  84  94  84 111 101
#>
#> $sample_2
#> [1] 114 111 114 114  84
```

Each bootstrap sample has five values, but some original values appear more than once and others do not appear at all.

Definition 0.2 (Bootstrap distribution)**.** Let $\hat{\theta}$ be a statistic computed from the observed data. Draw B independent bootstrap samples, and compute the statistic on each one, giving $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$. The **bootstrap distribution** of $\hat{\theta}$ is the empirical distribution of $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$.
Source: Wikipedia contributors [5].

Remark (What the bootstrap distribution estimates)

Remark 0.2 (What the bootstrap distribution estimates). The bootstrap distribution estimates the sampling distribution of $\hat{\theta}$. The observed sample stands in for the population, and resampling from it stands in for drawing new samples from the population.

Definition 0.3 (Bootstrap standard error)**.** The **bootstrap standard error** of a statistic $\hat{\theta}$, written $\widehat{\mathrm{SE}}_{\mathrm{boot}}$, is the sample standard deviation of the bootstrap replicates $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$ (Definition 0.2). It estimates the standard error of $\hat{\theta}$.
Source: Wikipedia contributors [6].

Example 0.2 (Bootstrap distribution of the sample mean of ten glucose values)**.** Take the baseline fasting glucose values of the first ten HERS participants, and compute the sample mean of each of $B = 1{,}000$ bootstrap samples:

```r
glucose10 <- hers$glucose[1:10]
set.seed(42)
boot_means10 <- replicate(1000, mean(sample(glucose10, replace = TRUE)))
c(
  observed_mean = mean(glucose10),
  bootstrap_se = sd(boot_means10),
  formula_se = sd(glucose10) / sqrt(10),
  divide_by_n_se = sqrt(mean((glucose10 - mean(glucose10))^2) / 10)
)
#>  observed_mean   bootstrap_se     formula_se divide_by_n_se
#>      103.80000        3.47992        3.61417        3.42870
```

The bootstrap standard error (Definition 0.3) is close to the usual estimate $s/\sqrt{n}$. Each bootstrap draw has variance $\hat{\sigma}^2 \overset{\text{def}}{=} \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$ around $\bar{x}$, and the n draws are independent, so as B grows the bootstrap standard error of the sample mean approaches $\hat{\sigma}/\sqrt{n}$ (`divide_by_n_se`), which is smaller than $s/\sqrt{n}$ by the factor $\sqrt{(n-1)/n}$. Figure 1 shows the bootstrap distribution.

```r
tibble::tibble(mean = boot_means10) |>
  ggplot2::ggplot() +
  ggplot2::aes(x = mean) +
  ggplot2::geom_histogram(bins = 30, color = "white") +
  ggplot2::geom_vline(xintercept = mean(glucose10), color = "red") +
  ggplot2::labs(x = "Bootstrap mean (mg/dL)", y = "Count")
```

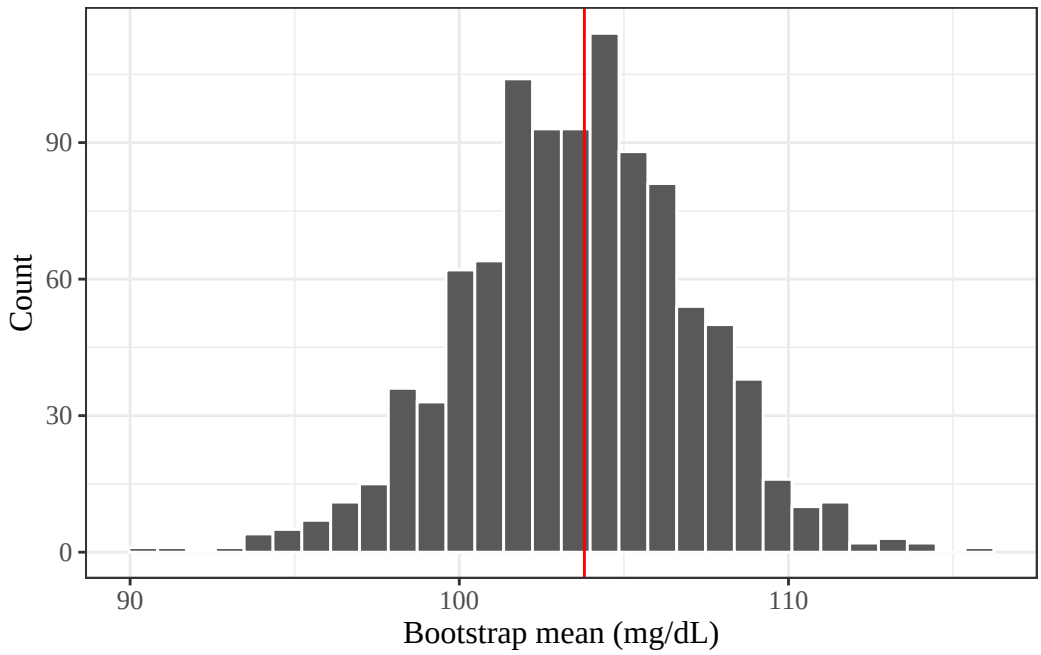


Figure 1: Bootstrap distribution of the sample mean fasting glucose of the first ten HERS participants ($B = 1{,}000$). The red line marks the observed sample mean.

Read

- Hogg et al. [7, Sec. 7.7], pp. 353–360 covers resampling methods.
- Casella and Berger [8, Sec. 10.1.4], pp. 478–480 covers bootstrap standard errors.
- Rice [9, Sec. 10.4.6], pp. 399–400 covers variability of location estimates by the bootstrap.
- Rice [9, Sec. 14.6], pp. 585–586 covers conditional inference, unconditional inference and the bootstrap.

Bootstrap confidence interval methods

There are three common methods for turning a bootstrap distribution into a $100(1-\alpha)\%$ confidence interval for a parameter θ [1, Ch. 3], [3]. Each is an approximate confidence interval. Throughout, z_q is the q quantile of the standard Gaussian distribution and Φ is its CDF.

Normal approximation

Definition 0.4 (Normal bootstrap confidence interval)**.** The **normal bootstrap confidence interval** is

$$\hat{\theta} \pm z_{1-\alpha/2}\, \widehat{\mathrm{SE}}_{\mathrm{boot}},$$

where $\widehat{\mathrm{SE}}_{\mathrm{boot}}$ is the bootstrap standard error (Definition 0.3).

ℹ Remark (When the normal bootstrap interval is reliable)

Remark 0.3 (When the normal bootstrap interval is reliable). This interval assumes that the sampling distribution of $\hat{\theta}$ is approximately Gaussian and centered at θ, so it can be unreliable when that distribution is skewed. It needs only a standard error, which takes fewer bootstrap replicates to estimate well than the tail quantiles used by the percentile and BCa intervals [3].

Example 0.3 (Normal bootstrap interval for the mean of ten glucose values)**.** Continuing Example 0.2, the 95% interval of Definition 0.4 is:

```
mean(glucose10) +
  c(lower = -1, upper = 1) * qnorm(0.975) * sd(boot_means10)
#>    lower    upper
#>  96.9795 110.6205
```

Percentile method

Definition 0.5 (Percentile bootstrap confidence interval)**.** The **percentile bootstrap confidence interval** runs from the $\alpha/2$ quantile to the $1-\alpha/2$ quantile of the bootstrap replicates $\hat{\theta}^*_1, \ldots, \hat{\theta}^*_B$ (Definition 0.2).
Source: Wikipedia contributors [10].

Remark (Number of bootstrap replicates for percentile intervals)

Remark 0.4 (Number of bootstrap replicates for percentile intervals). The extreme quantiles of B replicates are noisy estimates, so percentile-based intervals need more replicates than the normal interval; B of at least 1,000 is a common choice.

Example 0.4 (Percentile bootstrap interval for the mean of ten glucose values)**.** Continuing Example 0.2, the 95% interval of Definition 0.5 is:

```
quantile(boot_means10, c(0.025, 0.975))
#>  2.5% 97.5%
#>  96.5 110.4
```

Bias-corrected and accelerated method

Definition 0.6 (Bias correction of the BCa interval)**.** For a statistic $\hat{\theta}$ and its bootstrap replicates $\hat{\theta}^*_1, \dots, \hat{\theta}^*_B$, the **bias correction** is

$$\hat{z}_0 \stackrel{\text{def}}{=} \Phi^{-1}\left(\frac{\#\left\{b : \hat{\theta}^*_b < \hat{\theta}\right\}}{B}\right),$$

where Φ is the standard Gaussian CDF.

Example 0.5 (Bias correction when 40% of replicates fall below the estimate)**.** If 400 of $B = 1{,}000$ replicates are below $\hat{\theta}$, then $\hat{z}_0 = \Phi^{-1}(0.4) \approx -0.253$. If exactly half were below,

$$\begin{aligned}\hat{z}_0 &= \Phi^{-1}(0.5)\\ &= 0.\end{aligned}$$

Definition 0.7 (Acceleration of the BCa interval)**.** Let $\hat{\theta}_{(i)}$ be the statistic computed with observation i deleted, and let $\hat{\theta}_{(\cdot)}$ be the mean of $\hat{\theta}_{(1)}, \dots, \hat{\theta}_{(n)}$. The **acceleration** is

$$\hat{a} \stackrel{\text{def}}{=} \frac{\sum_{i=1}^{n}\left(\hat{\theta}_{(\cdot)} - \hat{\theta}_{(i)}\right)^3}{6\left(\sum_{i=1}^{n}\left(\hat{\theta}_{(\cdot)} - \hat{\theta}_{(i)}\right)^2\right)^{3/2}}.$$

Example 0.6 (Acceleration from three deleted estimates)**.** If the differences $\hat{\theta}_{(\cdot)} - \hat{\theta}_{(i)}$ are 1, 1, and -2, the sum of their cubes is $1+1-8=-6$ and the sum of their squares is 6, so

$$\hat{a} = \frac{-6}{6 \cdot 6^{3/2}} \approx -0.068.$$

Definition 0.8 (Bias-corrected and accelerated bootstrap confidence interval)**.** With the bias correction $\hat{z}_0$ and the acceleration $\hat{a}$, for $q \in \{\alpha/2, 1-\alpha/2\}$, let

$$\alpha_q \stackrel{\text{def}}{=} \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_q}{1 - \hat{a}(\hat{z}_0 + z_q)}\right).$$

The **bias-corrected and accelerated (BCa) bootstrap confidence interval** runs from the $\alpha_{\alpha/2}$ quantile to the $\alpha_{1-\alpha/2}$ quantile of the bootstrap replicates [3, Ch. 14].

ℹ Remark (What the BCa interval corrects)

Remark 0.5 (What the BCa interval corrects). When $\hat{z}_0 = 0$ and $\hat{a} = 0$, $\alpha_q = q$ and the BCa interval is the percentile interval (Definition 0.5). $\hat{z}_0$ measures how far the bootstrap distribution's median sits from $\hat{\theta}$, and $\hat{a}$ measures its skewness, so the BCa interval shifts the percentile interval to correct for both.

Example 0.7 (BCa bootstrap interval for the mean of ten glucose values)**.** Continuing Example 0.2, the quantities of Definition 0.8, step by step:

```
z0 <- qnorm(mean(boot_means10 < mean(glucose10)))
jackknife_means <- vapply(1:10, \(i) mean(glucose10[-i]), numeric(1))
dev <- mean(jackknife_means) - jackknife_means
a_hat <- sum(dev^3) / (6 * sum(dev^2)^(3 / 2))
z_q <- qnorm(c(0.025, 0.975))
alpha_q <- pnorm(z0 + (z0 + z_q) / (1 - a_hat * (z0 + z_q)))
c(z_0 = z0, a_hat = a_hat, alpha_lower = alpha_q[1], alpha_upper = alpha_q[2])
#>         z_0       a_hat alpha_lower alpha_upper
#>  0.01002668 -0.00867722  0.02422093  0.97422713
quantile(boot_means10, alpha_q)
#> 2.422093% 97.42271%
#>      96.5     110.4
```

The acceleration $\hat{a}$ is nearly 0 here, and $\hat{z}_0$ is small, so the BCa interval is close to the percentile interval of Example 0.4.

The boot package

The `boot` package [11], a recommended package distributed with R, provides `boot::boot()` to draw the bootstrap replicates and `boot::boot.ci()` to compute the three intervals. `boot::boot.ci()` differs from the definitions on this page in two small ways:

- its normal interval (`type = "norm"`) is centered at $\hat{\theta}$ minus the bootstrap estimate of bias, $\hat{\theta} - (\bar{\theta}^* - \hat{\theta})$, where $\bar{\theta}^*$ is the mean of the replicates, rather than at $\hat{\theta}$;
- it estimates quantiles of the replicates by interpolation, so its percentile and BCa endpoints can differ slightly from those computed with `quantile()`.

Example: slope of SBP on age in HERS

Example 0.8 (Bootstrap confidence intervals for the slope of SBP on age in HERS)**.** We regress systolic blood pressure (`SBP`) on `age` by ordinary least squares, and bootstrap the slope, resampling participants adapted from [1, Ch. 3]. The statistic function takes the data and the row indices of one bootstrap sample:

```
slope_sbp_age <- function(data, indices) {
  fit <- lm(SBP ~ age, data = data[indices, ])
  coef(fit)[["age"]]
}

set.seed(42)
boot_result <- boot::boot(data = hers, statistic = slope_sbp_age, R = 1000)
boot_result
#>
#> ORDINARY NONPARAMETRIC BOOTSTRAP
#>
#>
#> Call:
#> boot::boot(data = hers, statistic = slope_sbp_age, R = 1000)
#>
#>
#> Bootstrap Statistics :
#>     original       bias    std. error
#> t1* 0.471728 -0.000635537   0.0536514
```

The bootstrap standard error is close to the model-based standard error from `lm()`, and the three bootstrap intervals are close to the model-based 95% confidence interval:

```r
fit_sbp_age <- lm(SBP ~ age, data = hers)
parameters::parameters(fit_sbp_age) |>
  parameters::print_md(digits = 3, ci_digits = 3)
```

Parameter	Coefficient	SE	95% CI	t(2761)	p
(Intercept)	103.629	3.596	(96.579, 110.680)	28.820	$< .001$
age	0.472	0.054	(0.366, 0.577)	8.787	$< .001$

```r
boot::boot.ci(boot_result, type = c("norm", "perc", "bca"))
#> BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
#> Based on 1000 bootstrap replicates
#>
#> CALL :
#> boot::boot.ci(boot.out = boot_result, type = c("norm", "perc",
#>     "bca"))
#>
#> Intervals :
#> Level      Normal             Percentile            BCa
#> 95%   ( 0.3672,  0.5775 )   ( 0.3607,  0.5811 )   ( 0.3624,  0.5824 )
#> Calculations and Intervals on Original Scale
```

All three intervals are similar here. When the bootstrap distribution is skewed, the intervals differ more, and the BCa interval, which corrects for skewness, is the better choice [3, Ch. 14].

Further reading

- Chapter 24 of Lange's textbook [12] covers permutation tests and the bootstrap, with an emphasis on computation.

References

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